FYP MEETING LOG

Co-Supervisor Meeting 2 of 2

Agent-Based Game Analysis in Unity Based Upon Curiosity and RND Exploration Using PPO and SAC
Student: Raza Ullah Khan (NUM-BSCS23-28) | Supervisor: Dr. Malik M Ali Shahid | Co-Supervisor: Mr. Abdul Rafay | Industrial Partner: Inspired Square

Meeting No.	05 (Co-Supervisor Meeting 2)
Meeting With	Co-Supervisor - Mr. Abdul Rafay
Date	Tuesday, 3 November 2026
Time	12:00 PM - 12:20 PM
Duration	20 minutes
Mode / Venue	Co-Supervisor's office, Namal University Mianwali
Attendees	Raza Ullah Khan (Student); Mr. Abdul Rafay (Co-Supervisor)
Milestone	FN-02 (Month 1, Weeks 3-4): Hummingbirds 3D Flight Area & Flower Setup

1. Purpose of the Meeting

To review the Unity project I set up after the last meeting, look at a first draft of the PPO training config, and talk about one version problem I found.

2. Agenda

1. Review of the new Unity project and package list.
2. First draft of the baseline PPO config.
3. Version problem: RND support in mlagents 0.16.1.
4. How results will reach Weights & Biases.

3. What We Discussed

- **Unity project.** The project opens in Unity 2019.3.15f1. ML-Agents 1.0.2-preview and the Universal Render Pipeline 7.3.1 are in the package manifest. The trainer answers 'mlagents-learn --help' correctly.
- **Config draft.** In this ML-Agents version all settings sit in one YAML file under the agent's behaviour name. Sir said the draft is fine for now; real tuning starts only after the agent works (FN-04).
- **Version problem.** I checked and found that mlagents 0.16.1 has built-in Curiosity but not RND. RND and the PyTorch trainers come in later ML-Agents releases. Our proposal mentions both RND and PyTorch.
- **Plan for it.** We will keep the current versions for setup and baselines because they are stable with this Unity version. Before the exploration stage (FN-06/FN-07) we decide between upgrading ML-Agents or adding our own RND bonus. Whatever we choose, all six conditions must run on the same version so the comparison stays fair.
- **W&B.** This trainer writes TensorBoard logs into the results folder. We will sync those logs to Weights & Biases so all runs appear on one dashboard.

4. Small Implementation Part

These are the parts of the project that we reviewed together.

```
// Packages/manifest.json (main entries)
"com.unity.ml-agents": "1.0.2-preview",
"com.unity.render-pipelines.universal": "7.3.1",
"com.unity.textmeshpro": "2.0.1",
"com.unity.test-framework": "1.1.14"
```

```
# config/trainer_config.yaml  -  DRAFT baseline PPO (not tuned yet)
Hummingbird:
    trainer: ppo
    batch_size: 2048
    buffer_size: 20480
    learning_rate: 3.0e-4
    beta: 5.0e-3
    epsilon: 0.2
    lambd: 0.95
    num_epoch: 3
    hidden_units: 256
    num_layers: 2
    time_horizon: 128
    max_steps: 5.0e6
    summary_freq: 10000
    reward_signals:
        extrinsic:          # normal game reward only (baseline)
            strength: 1.0
            gamma: 0.99
        # curiosity:        # added later for Conditions 2 and 5
        #     strength: 0.02
        #     gamma: 0.99
```

Values are starting points taken from common ML-Agents examples. They will be tuned once and then kept the same for every condition.

5. Decisions Taken

- Keep Unity 2019.3.15f1 / ML-Agents 1.0.2-preview / mlagents 0.16.1 for the setup and baseline stage.
- Decide 'upgrade ML-Agents' vs 'own RND bonus' before FN-06; all six conditions must use one version.
- One shared YAML file; only the reward_signals part changes between conditions.
- Training logs will be synced to Weights & Biases.

6. Action Items

#	Action Item	Responsible	Due Date
1	Compare the ML-Agents release notes and list what changes if we upgrade.	Raza	17 Nov 2026
2	Keep the draft YAML on a feat branch until baseline tuning starts.	Raza	10 Nov 2026
3	Test syncing one dummy TensorBoard log to W&B.	Raza	10 Nov 2026
4	Share an example of a custom reward signal if we go that way.	Mr. Abdul Rafay	When needed

7. Other Notes

- The version issue will also be shared with the supervisor in the next meeting.

8. Next Meeting

Technical check-in once the agent's observations and actions work (FN-03, late November 2026).